

Monday Tutorial: Unit 4, Complex Impedance

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1 Memory Bank

- $V_{\text{rms}} = I_{\text{rms}} Z_{\text{tot}}$... Ohm's Law for AC circuits.
 - $P_{\text{ave}} = I_{\text{ave}} V_{\text{ave}} \cos \phi$... Average power in series RLC circuit.
 - $X_C = 1/(2\pi fC)$... Reactance of a capacitor.
 - $X_L = 2\pi fL$... Reactance of an inductor.
 - $Z_{\text{tot}} = \sqrt{R^2 + (X_L - X_C)^2}$... The total impedance of an RLC series circuit.
 - $\cos \phi = R/Z_{\text{tot}}$... Power factor of an RLC series circuit.
2. The impedance of an inductor is $Z_L = j\omega L$. Show that the reactance is just the magnitude (hypotenuse) of Z_L .
3. Consider the total impedance of an RLC circuit: $Z_{\text{tot}} = Z_R + Z_L + Z_C$. Show that the magnitude (hypotenuse) of Z_{tot} is the formula given in the memory bank.

2 Complex Numbers

1. Suppose that $j = \sqrt{-1}$, and that $z_1 = 2 + 4j$, and $z_2 = 2 - 2j$. The parts of these numbers not proportional to j are the *real parts*, and the *imaginary parts* are proportional to j . Numbers like z_1 and z_2 are called *complex numbers*. Calculate the following: (a) $z_1 + z_2$ (b) $z_1 - z_2$ (c) $z_1 * z_2$.
2. Suppose that $z_1 = 2 + 4j$, and $z_1^* = 2 - 4j$. That is, multiply the *imaginary part* by -1. (a) Calculate $z_1 * z_1^*$. (b) Let $z = x + jy$. Calculate $|z|^2 = zz^*$. What do you notice? (c) Graph a generic complex number, and note the hypotenuse and phase angle between the real and imaginary axes.
4. For an RLC series circuit having a 40.0Ω resistance, a 3.00 mH inductance, a $5.00 \mu\text{F}$ capacitance, and a voltage source with a V_{rms} of 120 V : (a) calculate the *phase angle*, ϕ , and *power factor*, $\cos \phi$, for $f = 60 \text{ Hz}$. (b) What is the average power at 60.0 Hz ? (c) Find the average power at the resonance frequency.
5. Considering what we know about complex numbers, and the solution to the previous exercise, prove that

$$P_{\text{ave}} = I_{\text{ave}} V_{\text{ave}} \cos \phi \quad (1)$$

3 Reactance and Impedance

1. The impedance of a capacitor is $Z_C = 1/(j\omega C)$. Show that the reactance is just the magnitude (hypotenuse) of Z_C .